Did the mutation come first?

Two hypotheses about where antibiotic resistance comes from, one square of sterile velvet, and the single statistic that separates them.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/mutation-first/

1 Before you open anything

Answer from what you already know. Getting it right is not the point — having a number on paper to argue with later is.

PREDICTION

Bacteria are grown into 36 separate colonies on a plate holding **no antibiotic**. A square of sterile velvet is pressed onto that plate and then onto three fresh plates that all **do** contain antibiotic, so the velvet stamps the same colony pattern three times. A few colonies grow on each of the three.

Take every grid position that grew a resistant colony on *at least one* of the three plates. **What percentage of those positions do you expect to have grown a resistant colony on all three plates?** Write one number between 0 and 100.

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In one or two sentences, say what your number assumes about *when* a cell becomes resistant.

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2 Reading one experiment

Open the demonstration. The top row of plates is what the velvet actually produced. The two rows below it are what each hypothesis predicts for that same run, computed from the same settings. The table headed **What the two hypotheses predict** is where you take your numbers from.

1. In the **Settings** panel, press **Reset everything**. Check that **Master plate grid** reads **8 × 6**, **Resistance frequency** reads **10%**, and **Velvet transfer misses** reads **0%**.
2. Record the run seed and the colony count from the line under **Run the experiment**, and the three readouts at the bottom of **Settings**, in the first table.
3. Copy all three rows of the comparison table into the second table, exactly as printed.

4. Tick **Reveal the master plate genotypes** and read the caption under the master plate. Record how many colonies it says already carry the mutation. Untick it again.

Run seed		Colonies on the master plate	
Expected resistant colonies per replica plate		Colonies already carrying the mutation	
Predicted agreement, antibiotic		Predicted agreement, mutation first	

Plates	**Resistant colonies #1 / #2 / #3**	**On all 3**	**On ≥1**	**Agree**
Observed				
Antibiotic causes it				
Mutation came first				

a. Compare the **Resistant colonies** column down all three rows. Could you tell the two hypotheses apart from those counts alone? Say why.

b. Which column *does* tell them apart, and what does that column count? Use the definition of **Agree** printed under the table.

c. Before you ticked the reveal box, nothing on the master plate image distinguished one colony from another. State what each hypothesis claims about that plate — how many resistant colonies is it carrying, according to each?

> The reveal box is not something a real experimenter has. On a plate with no antibiotic on it, a colony carrying the resistance mutation grows exactly like the rest, which is the reason the question has to be settled by transferring colonies rather than by looking at them.

3 What the frequency does and does not control

One slider, moved across its whole range. The three readouts at the bottom of **Settings** are calculated, not sampled, so they do not change when you press **New run**. The last column does come from a single run.

1. Keep **Master plate grid** at **8 × 6** and **Velvet transfer misses** at **0%** throughout.
2. Drag **Resistance frequency** to **3%**, its lowest setting. Record the three readouts, then the **Agree** figure on the **Observed** row.
3. Repeat at each frequency in the table, up to **25%**.

Resistance frequency	Expected resistant colonies per replica plate	Predicted agreement, antibiotic	Predicted agreement, mutation first	Observed row, Agree
3%				
5%				
10%				
15%				
20%				
25%				

> At **3%** the expected yield is about one colony per plate, so a run where nothing grows at all is ordinary. The **Agree** cell then reads a dash, because there is nothing to divide. Press **New run** until something grows, and note in the margin how many presses that took.

a. Which column changes steeply from top to bottom of your table? Which columns barely move, or do not move at all?

b. In your own words: what does the resistance frequency set, and what does it leave unchanged? One sentence, no numbers.

c. Go back to your Part 1 prediction. Which of the two predicted agreements is it closer to? If it was closer to the antibiotic row, name the assumption behind it; if it was closer to the mutation-first row, name the assumption too. Either way, write the assumption, not just the verdict.

4 When the velvet is imperfect

Real velvet does not lift every colony every time, and a prediction of exactly 100% agreement is not something any real plate delivers. This part is where a tempting shortcut breaks.

1. Press **Reset everything**, then set **Velvet transfer misses** to **20%**. Leave the frequency at **10%**.
2. Record both predicted agreements from the **Settings** readouts.
3. Press **New run** three times, recording the **Observed** row's **Agree** each time.

Predicted agreement, antibiotic	Predicted agreement, mutation first	Observed, run 1	Observed, run 2	Observed, run 3

a. Divide the mutation-first prediction by the antibiotic prediction. Roughly how many times larger is it? Show the division.

b. Here is the shortcut: *if the resistant colonies are not in identical positions on all three plates, the mutation cannot have come first.* Use your row of numbers to explain what is wrong with it, and say what the mutation-first hypothesis actually predicts when the velvet misses colonies.

c. Working backwards. Without touching the slider yet, write down the **Velvet transfer misses** value you think produces a **Predicted agreement, mutation first** closest to **61.6%**. Then find it on the slider and record what it actually is.

d. Compare the **Expected resistant colonies per replica plate** now with the value you recorded in Part 2, when the misses were at 0%. Explain in one sentence why raising the miss rate lowers that number, and why that drop is *not* what makes the two predicted agreements differ.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, set the controls the problem describes, and check yourself — and where you were wrong, write down which step went off.

1. The master plate grid is **8 × 6**, which puts 36 colonies on the dish. The resistance frequency is **25%** and the velvet misses **0%**. Predict the **Expected resistant colonies per replica plate** readout to one decimal place.

2. Now the grid is **10 × 8**, which puts 68 colonies on the dish; the frequency is **10%** and the velvet misses **25%** of colonies on every press.

(a) Predict the expected resistant colonies per replica plate.

(b) Under the mutation-first hypothesis a given mutant colony reaches all three plates with probability $(1 - m)^3$ and reaches at least one with probability $1 - m^3$, where m is the miss rate. Predict **Predicted agreement, mutation first** to one decimal place.

3. A lab reports that each of its three replica plates grew five resistant colonies, and concludes from the matching counts that the mutations were present before the antibiotic. State what is missing from that argument, and name the one additional piece of information that would settle it.

4. Joshua and Esther Lederberg (1952) were able to pick a resistant colony off a master plate that had never held any antibiotic, and grow a resistant culture from it. Explain how they knew which colony on that plate to pick, and why a culture grown from it rules out the antibiotic-causes-it hypothesis. Write your answer without saying that the bacteria adapted, responded, or needed anything.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- The **Many runs, not one lucky one** panel pools experiments instead of judging one. Press **Clear tally**, then **Run 25 more** several times, and watch the two mean agreements. Which one settles down fastest, and does either one drift toward the other?
- Use the **Both / Antibiotic causes it / Mutation first** buttons to dim one prediction, then press **Replay the stamp** and watch the three plates fill in order. Doing this with the mutation-first row hidden is a good test of whether you can call the pattern before it appears.
- Read the **Honest limits** panel. It puts the real mutation rate near one cell in 10^7 to 10^9, not the 10% the slider starts at. Work out roughly how many colonies a master plate would need before one resistant colony is likely — then decide whether that changes any conclusion you reached above, or only the number of plates required to reach it.